BUSINESS STATEMENT

Target is a globally renowned brand and a prominent retailer in the United States. Target makes itself a preferred shopping destination by offering outstanding value, inspiration, innovation and an exceptional guest experience that no other retailer can deliver.
This particular business case focuses on the operations of Target in Brazil and provides insightful information about 100,000 orders placed between 2016 and 2018. The dataset offers a comprehensive view of various dimensions including the order status, price, payment and freight performance, customer location, product attributes, and customer reviews.
By analyzing this extensive dataset, it becomes possible to gain valuable insights into Target's operations in Brazil. The information can shed light on various aspects of the business, such as order processing, pricing strategies, payment and shipping efficiency, customer demographics, product characteristics, and customer satisfaction levels

DATASET:

The data is available in 8 csv files:

1. customers.csv
2. sellers.csv
3. order_items.csv
4. geolocation.csv
5. payments.csv
6. reviews.csv
7. orders.csv
8. products.csv

1. Import the dataset and do usual exploratory analysis steps like checking the structure & characteristics of the dataset:

1.Data type of all columns in the "customers" table.

```sql
SELECT
    column_name,
    data_type
FROM `scaler-488208.Target.INFORMATION_SCHEMA.COLUMNS`
WHERE table_name = 'customers'
```

Row	column_name	data_type
1	customer_id	STRING
2	customer_unique_id	STRING
3	customer_zip_code_prefix	INT64
4	customer_city	STRING
5	customer_state	STRING

INSIGHTS:

- ONLY 5 COLUMN EXISTS IN THE CUSTOMERS TABLE
- 80% OF DATA STRING TYPE AND ONLY 20% INT64 TYPE

2.Get the time range between which the orders were placed.

```sql
SELECT MIN(order_purchase_timestamp) AS START_DAY,
       MAX(order_purchase_timestamp) AS RECENT_DAY
FROM `scaler-488208.Target.orders`
```

Row	START_DAY	RECENT_DAY
1	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC

INSIGHTS:

- THE ORDERS DATASET COVER MORE THAN 2 YEAR YEAR PERIOD, FROM LATE Q3 2016 THROUGH EARLY Q4 2018
- WHICH SUPPORTS ANALYSIS FOR ENABLING THE IDENTIFICATION OF CYCLICAL CUSTOMER ENGAGEMENT PATTERNS.

3.Count the Cities & States of customers who ordered during the given period.

```
SELECT COUNT(DISTINCT C.customer_city) AS CITY_COUNT,
      COUNT(DISTINCT C.customer_state) AS STATE_COUNT
FROM `scaler-488208.Target.customers` C
INNER JOIN `scaler-488208.Target.orders` O
USING(CUSTOMER_ID)
```

Row	CITY_COUNT	STATE_COUNT
1	4119	27

INSIGHTS:

- OVER THE PERIOD OF 2 YEARS THE BUSINESS SUCCESSFULLY ENGAGED OVER 4119 CITIES AND 27 STATES
- WHICH INDICATES THAT THE BUSINESS RUN ON NATIONWIDE NOT ONLY LIMITED TO LOCALISED AND STRONG CUSTOMER SUPPORT BASE

RECOMMENDATIONS:

- IDENTIFY THE PEAK SEASONS,AND STATES WITH CITIES WHERE CONSUMPTION IS HIGH AND PURCHASE PATTERN

2. In-depth Exploration:

1.Is there a growing trend in the no. of orders placed over the past years?

```
Select format_date('%Y-%m', order_purchase_timestamp) as
order_yr_mt,
     count(order_id) as orders
from `scaler-488208.Target.orders`
group by 1
order by 1
```

Row	order_yr_mt	orders
1	2016-09	4
2	2016-10	324
3	2016-12	1
4	2017-01	800
5	2017-02	1780
6	2017-03	2682
7	2017-04	2404
8	2017-05	3700
9	2017-06	3245
10	2017-07	4026

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INSIGHTS:

- DATA SCREENING THAT STRONG POTENTIAL OF GROWTH OVER PERIOD OF TIME.
- TERMINAL AND EARLY PHASE OF YEAR SHOWING HIGH VOLUME OF ORDER PLACED

RECOMMENDATIONS:

- INVENTORY AVAILABILITY DURING THE PEAK SEASON.
- IN THE LAST 2 MONTHS DATA SHOWN SHARP DECLINE OF ORDERS BUT IN MY OPINION ITS SHOWN JUST BECAUSE INSUFFICIENT OF DATA.

2.can we see some kind of monthly seasonality in terms of the no. of orders being placed?

```
Select EXTRACT(MONTH FROM order_purchase_timestamp) as order_month,
     count(order_id) as orders
from `scaler-488208.Target.orders`
group by 1
order by 1
```

Row	order_month	orders
1	1	8069
2	2	8508
3	3	9893
4	4	9343
5	5	10573
6	6	9412
7	7	10318
8	8	10843
9	9	4305
10	10	4959

INSIGHTS:

- SHARP DECLINE IN ORDER VOLUME WAS OBSERVED IN THE MONTH OF SEPTEMBER COMPARED TO THE PREVIOUS MONTH,DROP OF NEARLY 50% TO 60% FROM THE PEAK VOLUME RECORDED IN AUGUST.
- A HIGH GROWTH PATTERN WAS DEMONSTRATED FROM THE MONTH OF FEBRUARY THROUGH AUGUST.

RECOMMENDATIONS:

CRITICAL POINT NECESSITATING CONSIDERATION IS THE RESTRICTED DATA AVAILABILITY FOR SEPTEMBER AND OCTOBER IN BOTH THE INITIAL YEAR (2016) AND THE FINAL YEAR (2018). SUBSTANTIAL DATA VOLUME IS OBSERVED EXCLUSIVELY IN 2017. THIS DATA CONSTRAINT, WHICH MAY STEM FROM THE TERMINAL MONTH STATUS AND INSUFFICIENT DATA AVAILABILITY COMPARED TO OTHER MONTHS, WARRANTS ACKNOWLEDGEMENT.

3.During what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon or Night)

```
SELECT

CASE

WHEN Extract (hour FROM order_purchase_timestamp) BETWEEN 0 AND 6
THEN 'Dawn'

WHEN Extract (hour FROM order_purchase_timestamp) BETWEEN 7 AND 12
THEN 'Morning'

WHEN Extract (hour FROM order_purchase_timestamp) BETWEEN 13 AND 18
```

```
THEN 'Afternoon'

WHEN Extract (hour FROM order_purchase_timestamp) BETWEEN 19 AND 23
THEN 'Night'

END AS DAY_TIME,

COUNT(DISTINCT order_id) AS ORDER_COUNT

FROM `scaler-488208.Target.orders`

group by 1
```

Row	time_of_day	counter
1	night	28331
2	dawn	5242
3	afternoon	38135
4	morning	27733

INSIGHTS:

BRAZILIAN CUSTOMERS PREDOMINANTLY PLACE THEIR ORDERS DURING THE AFTERNOON AND RELATIVELY EQUAL VOLUME OF ORDERS DURING THE MORNING AND NIGHT, WHILE GENERALLY ABSTAINING FROM PLACING ORDERS DURING THE DAWN HOURS.

3.Evolution Of E-commerce orders in the brazil region:

1.Get the month on month no. of orders placed in each state.

```
Select customer_state, extract(month from order_purchase_timestamp)
as order_month,
     count(order_id) as total_orders
```

```
from `scaler-488208.Target.orders` o
inner join `scaler-488208.Target.customers` c
on o.customer_id = c.customer_id
group by 1,2
order by 1,2
```

Row	customer_state	order_month	total_orders
1	AC	1	8
2	AC	2	6
3	AC	3	4
4	AC	4	9
5	AC	5	10
6	AC	6	7
7	AC	7	9
8	AC	8	7
9	AC	9	5
10	AC	10	6

INSIGHTS:

THE FIVE STATES WITH THE HIGHEST ORDER VOLUME ARE SAO PAULO,RIO DE JANEIRO,MINAS GERAIS, RIO GRANDE DO SUL, AND PARANA.

CONVERSELY, THE FIVE STATES WITH THE LOWEST ORDER VOLUME ARE RORAIMA, AMAPA, ACRE,AMAZONAS,AND RONDONIA.

2.How are the customers distributed across all the states?

```
Select customer_state,

     count(*) as total_cust

from `scaler-488208.Target.customers`

group by 1

order by 2 desc;
```

Row	customer_state	total_cust
1	SP	41746
2	RJ	12852
3	MG	11635
4	RS	5466
5	PR	5045
6	SC	3637
7	BA	3380
8	DF	2140
9	ES	2033
10	GO	2020

INSIGHTS:

THE MAJORITY OF CUSTOMERS ARE CONCENTRATED IN FIVE KEY STATES: SAO PAULO, RIO DE JANEIRO, MINAS GERAIS, RIO GRANDE DO SUL, AND PARANA. THESE FIVE STATES COLLECTIVELY ACCOUNT FOR APPROXIMATELY 70% TO 80% OF THE COMPANY'S CUSTOMER BASE.

RECOMMENDATIONS:

NEED DEEP ANALYSIS OF THE TOP FIVE AND BOTTOM FIVE PERFORMING STATES IS REQUIRED TO FACILITATE BETTER DECISION-MAKING. FURTHERMORE, A BROADER DATASET ENCOMPASSING A LONGER TIMEFRAME AND OTHER KEY ASPECTS SHOULD BE INCORPORATED TO EXPAND COMPANY'S CUSTOMER BASE.

4. Impact on Economy: Analyze the money movement by e-commerce by looking at order prices, freight and others.

1.Get the % increase in the cost of orders from year 2017 to 2018 (include months between Jan to Aug only)

```
with final as (
Select extract(year from order_purchase_timestamp) as year_dt,
     round(sum(payment_value),2) as total_cost
from `scaler-488208.Target.orders`
inner join `scaler-488208.Target.payments`
```

```
using(order_id)
where extract(year from order_purchase_timestamp) in (2017,2018)
and extract(month from order_purchase_timestamp) between 1 and 8
group by 1)

Select year_dt, total_cost,
     round(100* (lead(total_cost) over(order by year_dt) -
total_cost)/total_cost,2) as Perc_cost_inc
from final
```

Row	year_dt	total_cost	Perc_cost_inc
1	2017	3669022.12	136.98
2	2018	8694733.84	*null*

INSIGHTS:

NEARLY 137% RISE OF COST OF TOTAL TRANSACTION VOLUME

2.Calculate the Total & Average value of order price for each state.

```
select customer_state,
      round(sum(price),2) as total_price,
      round(sum(price)/count(distinct order_id),2) as avg_order_val
from `scaler-488208.Target.customers`
inner join `scaler-488208.Target.orders`
using(customer_id)

inner join `scaler-488208.Target.order_items`
using(order_id)
group by 1
order by 2 desc,3 desc
```

Row	customer_state	total_price	avg_ord_val
1	SP	5202955.05	125.75
2	RJ	1824092.67	142.93
3	MG	1585308.03	137.33
4	RS	750304.02	138.13
5	PR	683083.76	136.67
6	SC	520553.34	144.12
7	BA	511349.99	152.28
8	DF	302603.94	142.4
9	GO	294591.95	146.78
10	ES	275037.31	135.82
11	PE	262788.03	159.46
12	CE	227254.71	171.25
13	PA	178947.81	184.48
14	MT	156453.53	173.26
15	MA	119648.22	161.69
16	MS	116812.64	164.76
17	PB	115268.08	216.67

INSIGHTS:

- IN STATES SUCH AS SAO PAULO, RIO DE JANEIRO, AND MINAS GERAIS MORE AFFORDABLE ITEMS ARE POPULAR, AND CUSTOMERS FREQUENTLY PURCHASE THESE TYPES OF PRODUCTS.
- CONVERSELY, STATES LIKE PARANA, PARA, AMAPA, AND ALAGOAS SHOW A PREFERENCE FOR PREMIUM PRODUCTS.
- STATES SUCH AS PARAIBA AND CEARA DEMONSTRATE HIGH-VOLUME SALES AND A CONCENTRATION OF HIGH-SPENDING CUSTOMERS.

RECOMMENDATIONS:

- IN STATES SUCH AS PARAIBA AND CEARA, PROTECTION FROM COMPETITORS IS REQUIRED, NECESSITATING THE INITIATION OF PROGRAMS SUCH AS LOYALTY SCHEMES.
- STATES INCLUDING SAO PAULO, RIO DE JANEIRO, AND MINAS GERAIS POSSESS A VAST CUSTOMER BASE, REQUIRING A FOCUS ON OPERATIONAL EFFICIENCY AND REDUCED SHIPPING COSTS TO MAINTAIN PROFIT MARGINS.
- STATES SUCH AS PARANA, PARA, AMAPA, AND ALAGOAS REQUIRE TARGETED DIGITAL MARKETING AIMED AT PREMIUM BUYERS, ALONGSIDE EFFORTS TO EXPAND THE CUSTOMER BASE.

3.Calculate the Total & Average value of order freight for each state

```
select customer_state,
     round(sum(freight_value),2) as total_val,
     round(sum(freight_value)/count(distinct order_id),2) as
avg_order_val
from `scaler-488208.Target.customers`
inner join `scaler-488208.Target.orders`
using(customer_id)

inner join `scaler-488208.Target.order_items`
using(order_id)
group by 1
order by 2 desc,3 desc
```

Row	customer_state	total_val	avg_order_val
1	SP	718723.07	17.37
2	RJ	305589.31	23.95
3	MG	270853.46	23.46
4	RS	135522.74	24.95
5	PR	117851.68	23.58
6	BA	100156.68	29.83
7	SC	89660.26	24.82
8	PE	59449.66	36.07
9	GO	53114.98	26.46
10	DF	50625.5	23.82

INSIGHTS:

- IN STATES SUCH AS SAO PAULO, RIO DE JANEIRO,MINAS GERAIS,RIO GRANDE DO SUL FREIGHT VOLUME IS GROWING AND THE SUPPLY CHAIN HANDLING IT EFFICIENTLY.
- CONVERSELY RORAIMA,RONDONIA,ACRE,PIAUI SHIPPING FEWER ORDERS BUT PAYING A PREMIUM FOR EACH ONE.

RECOMMENDATIONS:

- IN STATES SUCH AS SAO PAULO, RIO DE JANEIRO,MINAS GERAIS,RIO GRANDE DO SUL WHERE NEED TO MAINTAIN CURRENT CARRIER AND CONFIDENTLY INVEST MORE IN SALES AND MARKETING
- IN STATES SUCH AS RORAIMA,RONDONIA,ACRE,PIAUI NEED AUDIT CARRIER RATES IMMEDIATELY AND IMPLEMENT STRICT RULES AGAINST SENDING PARTIALLY EMPTY TRUCKS.

5 . Analysis based on sales, freight and delivery time

1. Find the no. of days taken to deliver each order from the order's purchase date as delivery time.Also, calculate the difference (in days) between the estimated & actual delivery date of an order.Do this in a single query.

```
select
order_delivered_customer_date,order_purchase_timestamp,order_estimated_delivery_date,

date_diff(order_delivered_customer_date,order_purchase_timestamp,day) as time_to_deliver,

date_diff(order_delivered_customer_date,order_estimated_delivery_date,day) as diff_estimated_delivery
from `scaler-488208.Target.orders`
```

Row	order_delivered_customer_date	order_purchase_timestamp	order_estimated_delivery_date	time_to_deli...	diff_estimated_de...
1	2016-11-08 10:58:34 UTC	2016-10-03 21:01:41 UTC	2016-11-25 00:00:00 UTC	35	-16
2	2016-11-09 14:53:50 UTC	2016-10-09 15:39:56 UTC	2016-12-08 00:00:00 UTC	30	-28
3	2018-03-21 22:03:51 UTC	2018-02-19 19:48:52 UTC	2018-03-09 00:00:00 UTC	30	12
4	2016-11-09 07:47:38 UTC	2016-09-15 12:16:38 UTC	2016-10-04 00:00:00 UTC	54	36
5	2016-11-17 08:38:57 UTC	2016-10-04 15:44:44 UTC	2016-11-24 00:00:00 UTC	43	-6
6	2016-11-09 13:37:38 UTC	2016-10-04 12:53:17 UTC	2016-11-24 00:00:00 UTC	36	-14
7	2016-11-03 10:08:51 UTC	2016-10-04 16:05:29 UTC	2016-11-24 00:00:00 UTC	29	-20
8	2016-11-04 17:54:00 UTC	2016-10-04 18:52:56 UTC	2016-11-24 00:00:00 UTC	30	-19
9	2016-11-18 08:51:07 UTC	2016-10-04 09:59:03 UTC	2016-11-24 00:00:00 UTC	44	-5
10	2016-12-12 20:31:54 UTC	2016-10-04 21:54:52 UTC	2016-11-24 00:00:00 UTC	68	18

INSIGHTS:

- A FOCUS ON OPERATIONAL EFFICIENCY IS REQUIRED DUE TO DATA FINDINGS INDICATING EXTENDED DELIVERY TIMES, OCCASIONALLY EXCEEDING 50 DAYS. FURTHERMORE, A SUBSTANTIAL VOLUME OF DATA REVEALED ISSUES SUCH AS INCOMPLETE SCREENING, OBSCURE VALUES, AND A NEED FOR IMPROVED FORECASTING REGARDING ESTIMATED DATES.

2.Find out the top 5 states with the highest & lowest average freight value

```
with highest as
(Select c.customer_state,
     round(avg(freight_value),2) as avg_freight_value,
     'Highest' as status
from `scaler-488208.Target.orders` o
inner join `scaler-488208.Target.customers` c
on o.customer_id = c.customer_id

inner join `scaler-488208.Target.order_items` i
on i.order_id = o.order_id
group by 1
order by 2 desc
limit 5 ),

lowest as
(Select c.customer_state,
     round(avg(freight_value),2) as avg_freight_value,
     'Lowest' as status
from `scaler-488208.Target.orders` o
inner join `scaler-488208.Target.customers` c
on o.customer_id = c.customer_id

inner join `scaler-488208.Target.order_items` i
```

```
on i.order_id = o.order_id
group by 1
order by 2
limit 5 )

select * from highest
union distinct
select * from lowest
```

Row	customer_state	avg_freight_value	status
1	RR	42.98	Highest
2	PB	42.72	Highest
3	RO	41.07	Highest
4	AC	40.07	Highest
5	PI	39.15	Highest
6	SP	15.15	Lowest
7	PR	20.53	Lowest
8	MG	20.63	Lowest
9	RJ	20.96	Lowest
10	DF	21.04	Lowest

INSIGHTS:

THESE ARE TOP 5 STATES WHERE HIGHEST AVERAGE FREIGHT VALUE RORAIMA,PARAIBA,RONDONIA,ACRE,PIAUI

AND THESE ARE BOTTOM 5 STATES WHERE LOWEST AVERAGE FREIGHT VALUE SAO PAULO,PARANA,MINAS GERAIS,RIO DE JENEIRO,DISTRITO FEDERAL

3.Find out the top 5 states with the highest & lowest average delivery time

```
with highest as
(Select c.customer_state,
     round(avg(a.time_to_deliver),2) as average_delivery_time,
     'Highest' as status
from `scaler-488208.Target.orders` o
inner join `scaler-488208.Target.customers` c
on o.customer_id = c.customer_id
```

```
inner join (select
order_id,order_delivered_customer_date,order_purchase_timestamp,orde
r_estimated_delivery_date,

date_diff(order_delivered_customer_date,order_purchase_timestamp,day
) as time_to_deliver
from `scaler-488208.Target.orders`) a
on a.order_id = o.order_id
group by 1
order by 2 desc
limit 5 ),

lowest as
(Select c.customer_state,
     round(avg(a.time_to_deliver),2) as average_delivery_time,
     'Lowest' as status
from `scaler-488208.Target.orders` o
inner join `scaler-488208.Target.customers` c
on o.customer_id = c.customer_id

inner join (select
order_id,order_delivered_customer_date,order_purchase_timestamp,orde
r_estimated_delivery_date,
date_diff(order_delivered_customer_date,order_purchase_timestamp,day
) as time_to_deliver
from `scaler-488208.Target.orders`) a
on a.order_id = o.order_id
group by 1
order by 2
limit 5 )

select * from highest
```

```
union distinct
select * from lowest;
```

Row	customer_state	average_delivery_...	status
1	RR	28.98	Highest
2	AP	26.73	Highest
3	AM	25.99	Highest
4	AL	24.04	Highest
5	PA	23.32	Highest
6	SP	8.3	Lowest
7	PR	11.53	Lowest
8	MG	11.54	Lowest
9	DF	12.51	Lowest
10	SC	14.48	Lowest

INSIGHTS:

THESE ARE TOP 5 STATES WHERE HIGHEST AVERAGE DELIVERY TIME REQUIRED

RORAIMA,AMAPA,AMAZONAS,ALAGOAS,PARA

AND THESE ARE BOTTOM 5 STATES WHERE AVERAGE DELIVERY TIME REQUIRED

SAO PAULO,PARANA,MINAS GERAIS,DISTRITO FEDERAL,SANTA CATARINA

4.Find out the top 5 states where the order delivery is really fast as compared to the estimated date of delivery.You can use the difference between the averages of actual & estimated delivery date to figure out how fast the delivery was for each state

```
select
   c.customer_state,
   round(avg(date_diff(o.order_delivered_customer_date,
o.order_purchase_timestamp, DAY)), 2) as
average_actual_delivery_time,
   round(avg(date_diff(o.order_estimated_delivery_date,
o.order_purchase_timestamp, DAY)), 2) as avg_estimate_delivery,
   round(avg(date_diff(o.order_estimated_delivery_date,
o.order_delivered_customer_date, DAY)), 2) as
days_faster_than_estimate,
   'Top' as status
from `scaler-488208.Target.orders` o
```

```
inner join `scaler-488208.Target.customers` c
   using(customer_id)
where o.order_delivered_customer_date is not null
group by 1
order by 4
limit 5;
```

Row	customer_state	average_actual_d...	avg_estimate_deli...	days_faster_than...	status
1	AL	24.04	32.21	7.95	Top
2	MA	21.12	30.08	8.77	Top
3	SE	21.03	30.48	9.17	Top
4	ES	15.33	25.22	9.62	Top
5	BA	18.87	29.07	9.93	Top

INSIGHTS:

THESE ARE TOP 5 STATES WHERE DELIVERY IS REALLY FAST AS COMPARED TO ESTIMATED DELIVERY TIME PERIOD ALAGOAS,MARANHAO,SERGIPE,ESPIRITO SANTO,BAHIA

6 . Analysis based on the payments:

1.Find the month on month no. of orders placed using different payment types.

```
Select p.payment_type,extract(month from order_purchase_timestamp)
as month,
     count(o.order_id) as total_orders
from `scaler-488208.Target.orders` o
inner join `scaler-488208.Target.payments` p
on p.order_id = o.order_id
group by 1,2
order by 2
```

Row	payment_type	month	total_orders
1	debit_card	1	118
2	voucher	1	477
3	UPI	1	1715
4	credit_card	1	6103
5	voucher	2	424
6	credit_card	2	6609
7	UPI	2	1723
8	debit_card	2	82
9	credit_card	3	7707

INSIGHTS:

AS PER DATA MOST OF TRANSACTION OCCURED THROUGH CREDIT CARD AND UPI OTHERS ARE LOWER COMPARED TO THIS TWO MEDIUM

2.Find the no. of orders placed on the basis of the payment installments that have been paid.

```
Select p.payment_installments,
     count(o.order_id) as total_orders
from `scaler-488208.Target.orders` o
inner join `scaler-488208.Target.payments` p
on p.order_id = o.order_id
where p.payment_installments >=1
group by 1
order by 2 desc
```

Row	payment_installm...	total_orders
1	1	52546
2	5	5239
3	2	12413
4	3	10461
5	4	7098
6	10	5328
7	6	3920
8	8	4268
9	9	644
10	7	1626

INSIGHTS:

- MOST OF THE ORDERS PLACED WITH 1 INSTALLMENTS MAKING IT IS THE MOST PREFERRED OPTION
- AS THE NUMBER OF INSTALLMENTS INCREASES QUANTITY OF ORDER DECREASING INDICATING LONG INSTALLMENT IS NOT PREFERRED BY CUSTOMER

RECOMMENDATIONS:

- PROMOTE AND OPTIMIZE 1-2 INSTALLMENT PLAN AS MOST OF CUSTOMER PREFER IT.
- INCENTIVIZE HIGHER INSTALLMENT PLANS LIKE NO-COST EMI,DISCOUNTS,CASHBACKS ETC FOR ENCOURAGE TO EXPLORE THIS OPTION ALSO